Media Hash List – Quick Start Guide

Version 0.1.30

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Media Hash List – a digital packing list

In file-based workflows of digital cinematography today the secure and reliable transfer of media files is one of the essential activities to be performed in a production. The Media Hash List specification ensures that the complete source media is transferred without any alterations.

Media Hash List – files and tools

A Media Hash List file can be described as a packing list for a set of files in a folder structure.

The Media Hash List initiative provides tools that perform the necessary steps for

- sealing (creating Media Hash List files for a set if given files),
- and checking consistency and completeness (verifying a set of given files with a given Media Hash Lists file).

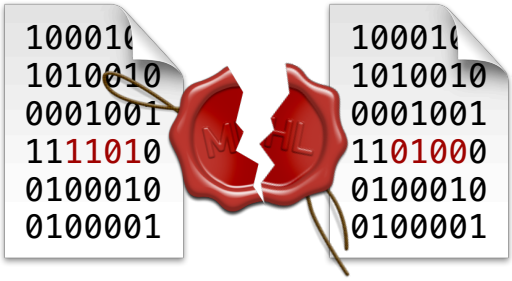
Sealing

The Media Hash List standard allows you to “seal” a filesystem folder.

By sealing a folder the creator of the seal affirms that the folder has been in a well-defined state. Any change to the folder breaks the seal and can therefore easily be detected. Media Hash List assures both the consistency of the folder contents as well as the completeness of the folder structure compared to the state when the seal was created.

Technically speaking, the MHL tool gathers the folder’s current state and store it to an XML-based Media Hash List file. This MHL file comprehends all the information necessary to tell if the folder structure or any of the contained files was modified since the MHL file was created.

Consistency

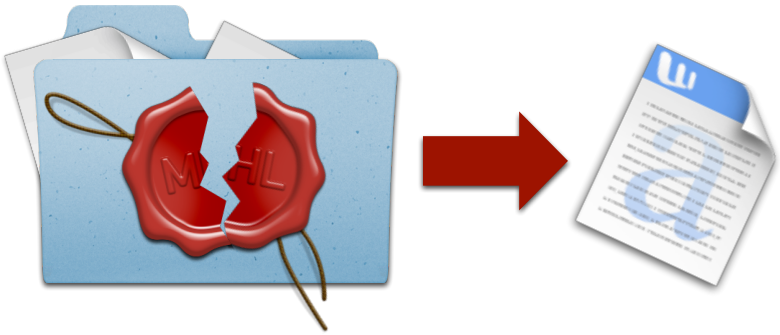


Consistency means that your files don't change over time. Consistency is especially important if your files get copied multiple times throughout their lifespan. You want to make sure that the copy is an exact replica of the original.

The best solution to this problem are checksums, which are commonly used in cryptography. Media Hash List utilizes the widely used SHA-1 and MD5 hashing algorithms. While being similar to cross sums, these algorithms are mathematically much more advanced and guarantee that no two distinct files possess the same checksum.

Therefore, you can detect any change in a file by comparing it to its checksum. If the file wasn't altered, the checksums match. If the file was altered, it is by definition another file and therefore, the checksums don't match.

Completeness



Completeness means that all of your files and their structure are preserved over time. Throughout their lifespan, your files get copied and moved multiple times. During each file operation a multitude of unintended changes can occur, mostly due to human error:

- Change of folder structure
- Incomplete copy of a folder structure
- Removal of files
- Addition of files

To detect these errors, Media Hash List stores a map of the original folder structure in each MHL file. Therefore, every change to the folder structure can be detected by comparing it to the original map.

First steps

- **Download**
  Download the command line tools for Media Hash List from
  http://mediahashlist.org/
- **Extract**
  Extract the downloaded ZIP file .
- **Install the binaries on your system**
  (Replace the <MHL_TOOLS_FOLDER> placeholder with the extracted folder path)
  - Windows:
    - In Explorer navigate to `<MHL_TOOLS_FOLDER>/Windows/`
    - Copy the `mhl.exe` to `C:\Windows\System32\`
  - Mac:
    - In the Finder navigate to `<MHL_TOOLS_FOLDER>/Mac/`
    - Double click the installer: `Install MHL Tool.pkg`
  - Linux
    - Open a shell
    - Type the following at the command prompt:
      - `cd <MHL_TOOLS_FOLDER>/Linux/`
      - `sudo cp mhl /usr/bin`
- **Check installation**
  On every system you now see an usage message from the MHL Tool if you type `mhl --version` (Linux and Mac) or `mhl.exe --version` (Windows) in a terminal window or prompt:

```
$ mhl --version
Version: 0.1.28 (Build: 736)
```

Examples

In the following we give a few examples how the `mhl` command can be used in conjunction to create and verify MHL files.

Creating a Media Hash List for contents of a folder

Change to the directory where the files are located and execute the following command. A new MHL file will be created.

```
$ mhl seal -t md5 *
```

Creating a Media Hash List for files FILE_A and FILE_B

Change to the directory where the files are located and execute the following command. A new MHL file will be created alongside the files.

```
$ mhl seal -t md5 FILE_A FILE_B
```

Verifying a MHL file

Change to the directory where the MHL file is located and execute the following command.

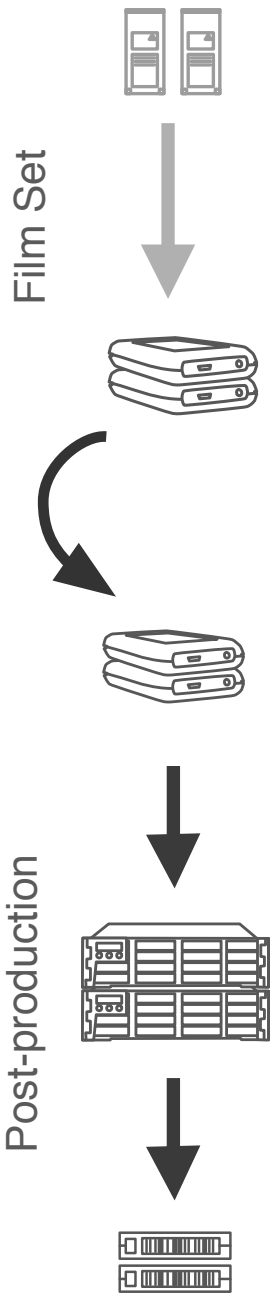
```
$ mhl verify -f FILE.mhl
```

Help and more usage information

More information is available with the help command:

```
$ mhl help
```

Example Media Hash List Production Workflow



Offload of source media

The very first MHL file should ideally be created by the copy tool you are using to offload your files from the source media.

A copy tool is the only way to assure that the offloaded folder is an exact copy of the original files and that no error occurred during the copy. The Media Hash List initiative is working to get vendors of copy tools on board to natively create MHL files.

In case your copy tool does not yet support Media Hash List, you can seal the offloaded folders manually. Thereby you affirm that the folder is in a well-defined state and can be sent off to post. However, this does not allow you to verify that the offloaded folder is the very same as the source

Arrival of transfer drive in Post

Once the transfer disks arrive in the post, you use the MHL file (e.g. with the MHL Tool) to verify that the files on the disk weren't damaged. This can happen during a disk head crash, bitflips due to magnetic fields, etc.

Backup in Post

When copying the files in the post make sure you copy the MHL file created on-set along with the files. It is advisable to check the completeness of the folders after every copy operation. Every now and then you should also check the consistency of the contained files.

Archive & Restore

When archiving folders make sure you copy all MHL files along with it. Check the completeness and the consistency after every restore.

MHL files

Media Hash List file structure

MHL files are a human-readable XML format. They contain information about each sealed file together with creator info. An XML Schema for Media Hash List is available at http://mediahashlist.org/.

Below is an example of a basic MHL file.

```
<?xml version="1.0" encoding="UTF-8"?>
<hashlist version="1.0">

    <creatorinfo>
        <name>Franz Klammer</name>
        <username>fkl</username>
        <hostname>workst10.local</hostname>
        <tool>mhl ver. 0.1.27</tool>
        <startdate>2013-07-15T09:22:39Z</startdate>
        <finishdate>2013-07-15T09:22:40Z</finishdate>
    </creatorinfo>

    <hash>
        <file>A001R1KL/A001C002_111011_R1KL.mov</file>
        <size>164257195</size>
        <lastmodificationdate>2012-08-23T10:51:20Z</lastmodificationdate>
        <md5>6eb41a9981060d311325cc266e654d40</md5>
        <hashdate>2013-07-15T09:22:39Z</hashdate>
    </hash>
    <hash>
        <file>A001R1KL/A001C003_111011_R1KL.mov</file>
        <size>166497735</size>
        <lastmodificationdate>2012-08-23T10:51:20Z</lastmodificationdate>
        <md5>03b17a4947f338bab9b25f82e1571373</md5>
        <hashdate>2013-07-15T09:22:39Z</hashdate>
    </hash>

    [...]

</hashlist>
```

Media Hash List file naming

The MHL tool names the created MHL file as follows:

<foldername>_<date>_<time>.mhl

where the placeholders are substituted with the following:

- <foldername> – name of the current directory, e.g. ‘Movies’
- <date> – current date in format YYYY-MM-DD, e.g. ‘2013-05-10’
- <time> – current time in format HHMMSS, e.g. ‘111527’

The current directory is the folder you were in when calling the MHL tool.

Media Hash List file location

The MHL tool requires the MHL file to be placed along the path of the referenced file. This allows for an easy discovery when looking for an MHL file for a specific file.

For example, when creating a MHL file for the file

```
/Example/Movies/Clip1.mov
```

the MHL file can be put in the following locations:

```
/Example/Movies/
/Example/
/
```

By default, the MHL file is located in the current directory you were in when calling the MHL tool (see section 'MHL file naming' above). You can manually specify the location of the MHL file, by passing the '`--output-path`' option to the MHL tool.